

Neutrophil-Like Cells Response to Experimental Metal Alloys



Paloma Zarazua¹, and Steve Alas¹

Department of Biological Sciences
California State Polytechnic University, Pomona¹

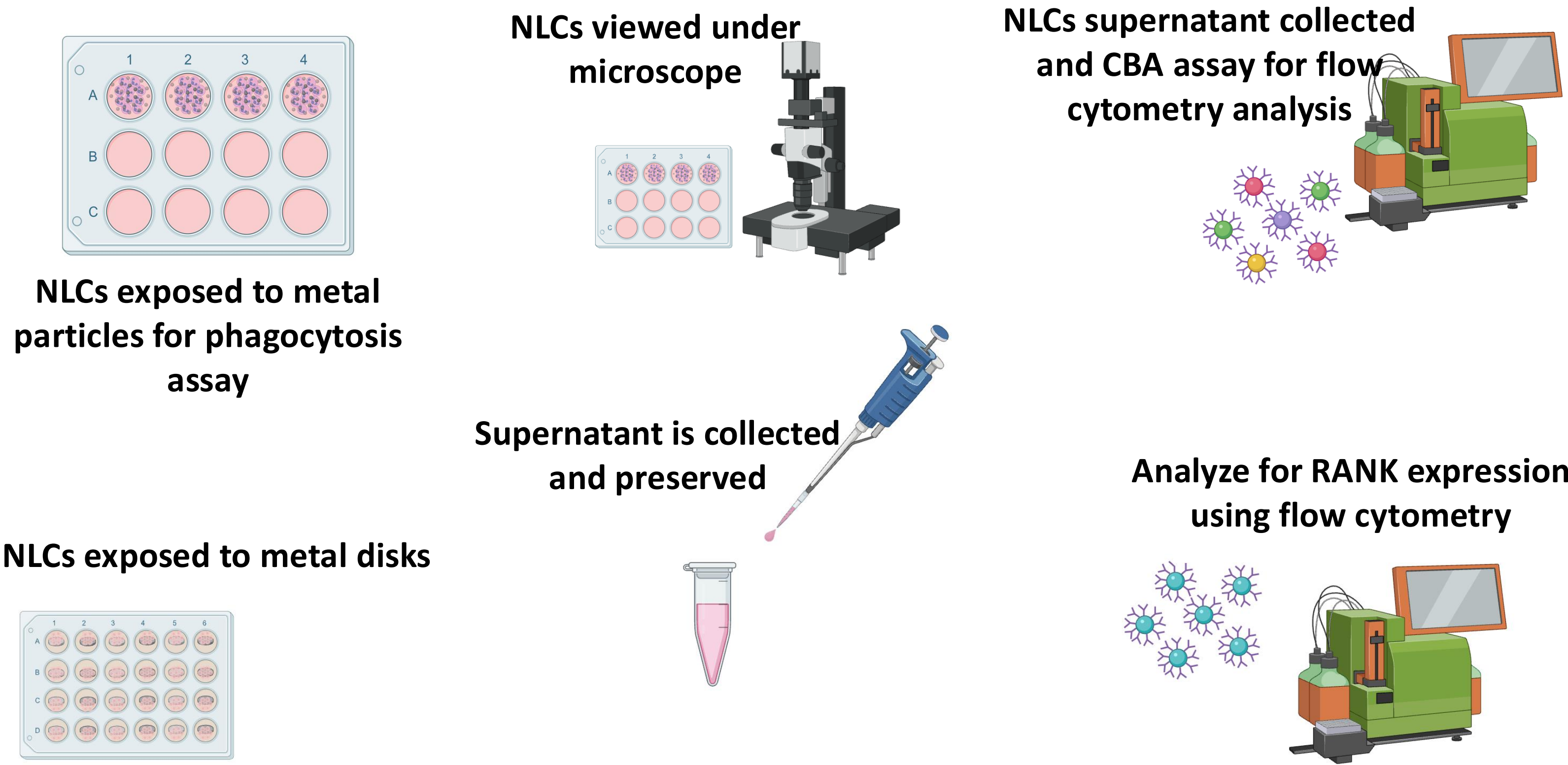


Abstract

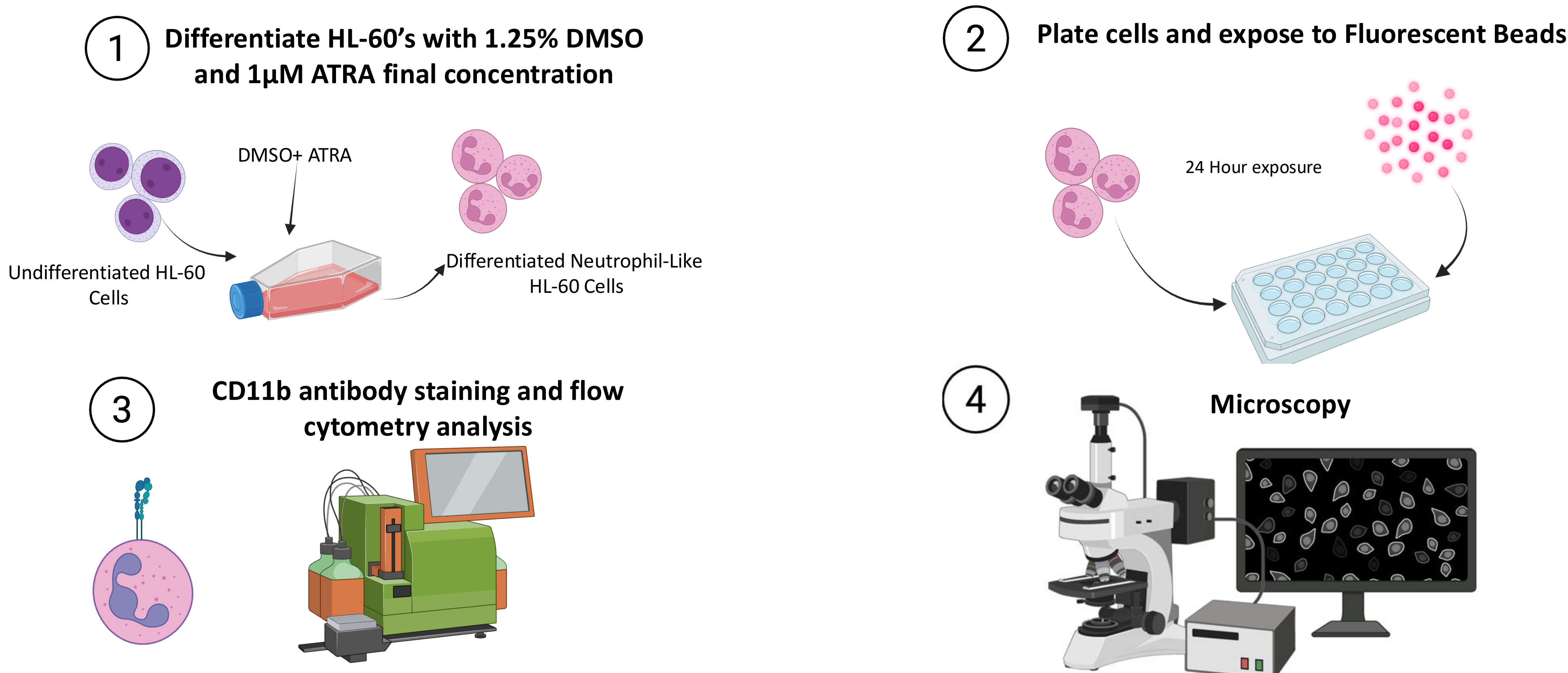
The demand for prosthetics has gradually increased with the advancements in technology, making them more accessible to the growing population in the United States. Prosthetic recipients have become increasingly younger over time, resulting in a longer duration of implant use among this demographic. The extended implant duration can result in corrosion of the metal alloy, creating an inflammatory response which can lead to loosening of the implant due to bone resorption. Bone resorption is the process of bone tissue dissolving through the action of specialized cells called osteoclasts. Monocytes and macrophages are precursors for osteoclast cells. One of the key signals in osteoclast formation is the Receptor Activator of Nuclear factor Kappa beta (NFkB ligand), also known as RANKL. This ligand plays a crucial role by binding to the RANK receptors found on the surface of monocytes/macrophages promoting differentiation of osteoclasts. Mature osteoclasts will attach to the bone surface and secrete hydrochloric acid that degrades the bone, causing osteolytic loosening. Different cell types are involved in the activation of bone resorption including osteoclasts, macrophages and fibroblasts, but it is not entirely known how neutrophils play a role in this process. Neutrophils are one of the first responders of the immune system in the presence of an antigen. They secrete many of the same cytokines as macrophages and fibroblasts that aid in osteoclast activation. It is unknown how neutrophils respond to the metals directly or if they induce an inflammatory response. This study seeks to provide more knowledge on neutrophil function in the presence of new metal alloys that could potentially be used for prosthetic implants. HL-60 cells differentiated into neutrophil-like cells (NLCs) will be exposed to titanium, titanium-aluminum-vanadium alloys (Ti-6Al-4V), and novel boron-titanium alloys. Upon exposure, phagocytosis activity will be analyzed along with inflammatory cytokine secretion using a cytometric bead array, and expression of RANKL by flow cytometry. This analysis will compare a series of metal alloys and determine which one activates neutrophils the least, making it the most biocompatible.

Future Aims

- ❖ Microscopy for Phagocytosis of different titanium metal particle concentrations such as Ti64, Ti64+ 0.5B, and Ti64 +1.0B.
- ❖ Measurement of IL-1 β , IL-6, TNF- α , IL-8, and GM-CSF released by NLCs after exposure to titanium particles and titanium metal disks using a CBS kit
- ❖ Flow cytometry analysis of RANK expression on NLCs upon exposure to titanium metal particles and titanium metal disks.



HL-60 Cell Differentiation Into Neutrophil-Like Cells (NLCs) and NLC Phagocytic Activity



Flow Cytometry Analysis

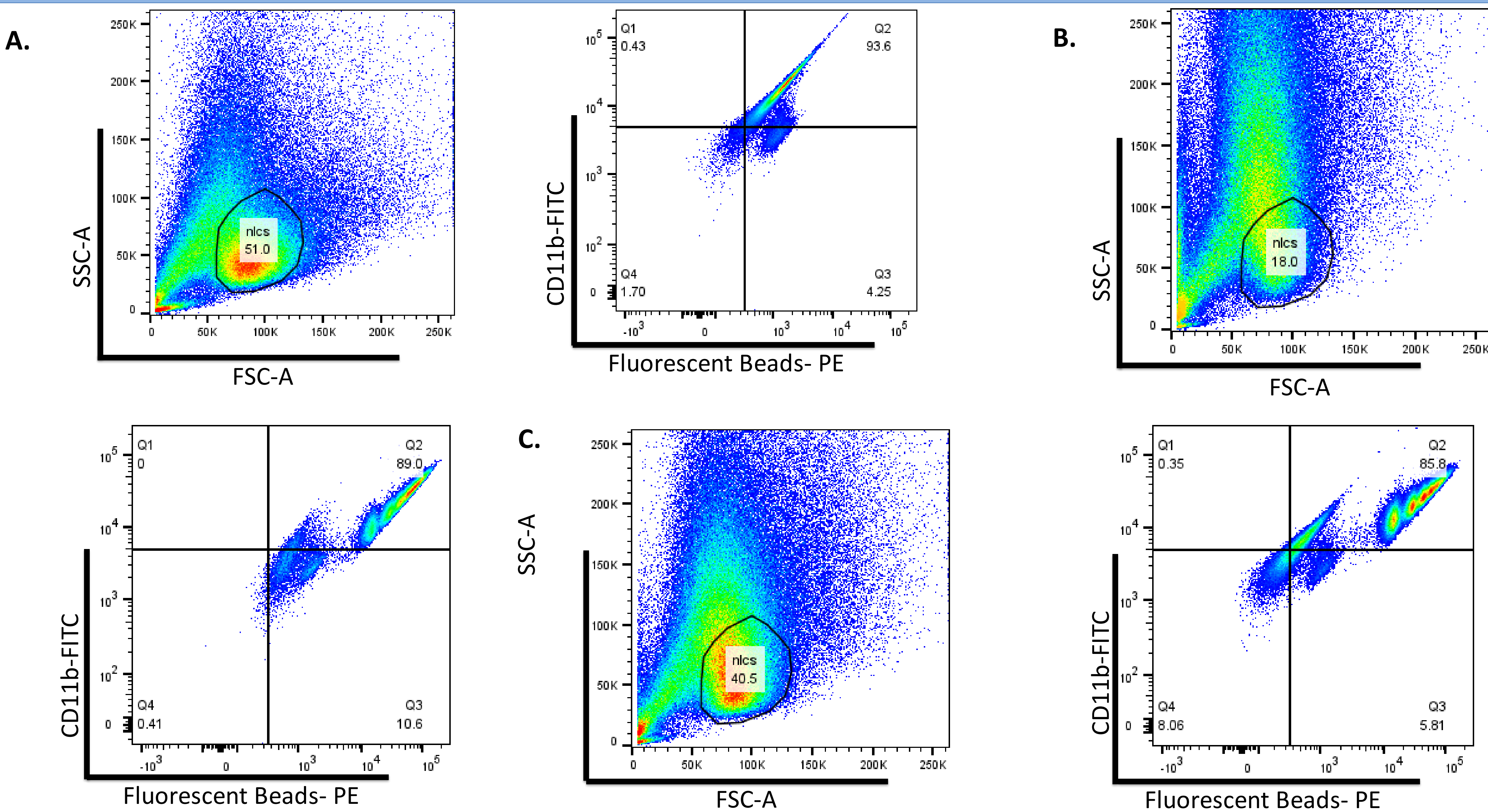


FIG 1: HL-60 Differentiation after 3 days and 24 hour incubation with fluorescent beads post differentiation flow cytometry analysis. The HL-60 cell line was differentiated into neutrophil-like cells using 1.25% DMSO and 1 μ M all-trans retinoic acid for 3 days. CD11Bb expression was analyzed in NLCs (A), NLCs with a 1:10 dilution of fluorescent beads for 24 hours (B), and NLCs with a 1:100 dilution of fluorescent beads for 24 hours.

NLCs Phagocytosis Microscopy of Fluorescent Beads

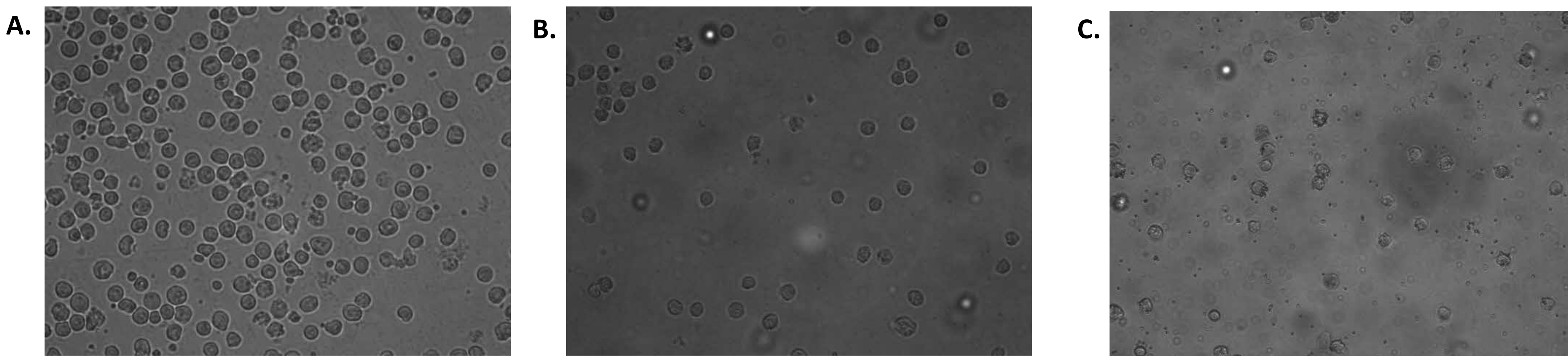


FIG 2: HL-60 Differentiation after 3 days and 24 hour incubation with fluorescent beads post differentiation Microscopy. Neutrophil-like cells were differentiated using the HL-60 cell line. Post differentiation, NLCs were exposed to fluorescent beads to view phagocytic activity. (A) HL-60 cells with no beads (B.) NLCs with no beads (C.) NLCs with a 1:10 dilution of fluorescent beads.

Acknowledgments

We thank the staff of the Biological Sciences Department at Cal Poly Pomona for their assistance in aiding resources for this project.